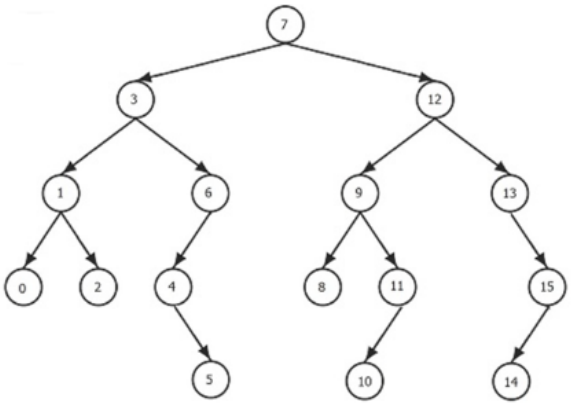
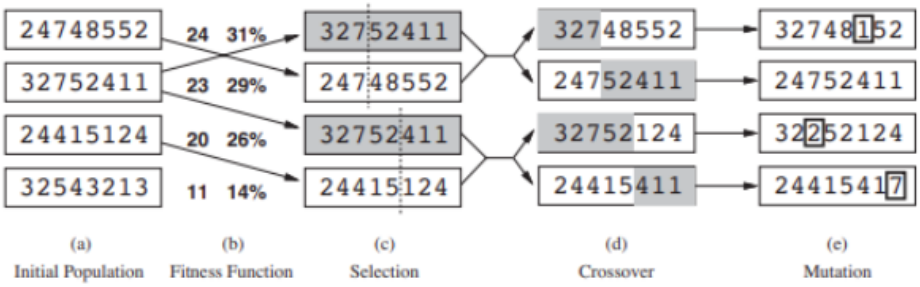
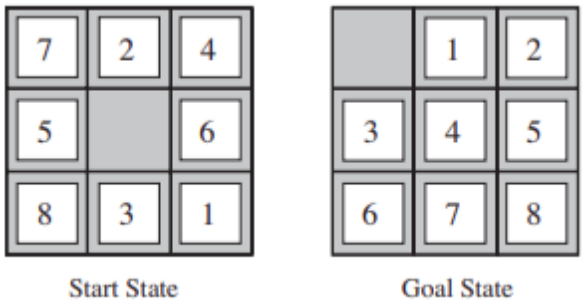
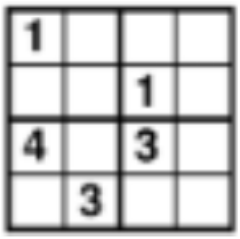
Q. No.	Questions
UNIT 1	**SECTION 1**
1	A multinational company is considering two AI deployment strategies: Strategy A: Process all data in centralized cloud servers. Strategy B: Process data directly on edge devices such as smartphones and IoT sensors. The company operates autonomous vehicles that require rapid decision-making. Compare the suitability of cloud-based AI and Edge AI for this application and recommend the most appropriate strategy with supporting reasons. (5 Marks)
2	A self-driving taxi receives complaints from passengers about sudden braking, uncomfortable turns, and abrupt acceleration during daily rides. To improve the travel experience, the system continuously records passenger feedback along with driving data such as speed, braking intensity, and steering patterns. Outline the roles of the critic and learning element in improving the taxi's performance and justify the contribution of feedback to make better future decisions. (5 Marks)
3	A hospital deploys an autonomous sterilization robot to disinfect patient rooms every night. For several weeks, the robot has repeatedly used the same cleaning sequence because it consistently completes all assigned rooms before the deadline. Recently, a new wing has been opened, some corridors are temporarily closed for maintenance, and updated elevator schedules have reduced waiting times on certain floors. Despite these changes, the robot continues using its old cleaning sequence without trying any alternatives. Contrast the responsibilities of the problem generator and the performance element in this learning agent and indicate an exploratory actions that the robot could perform during low-risk operating periods. (5 Marks)
4	(i) A hospital plans to implement an AI-based medical diagnosis system that assists doctors by recommending possible diseases based on patient symptoms.The system cannot directly observe the patient's internal health conditions. Resolve the task environment of the medical diagnosis system by classifying it with respect to observability, determinism, and agent type. (3 Marks)
	(ii) Integrate the relationship between the identified environmental characteristics and the design of a rational diagnostic agent. (2 Marks)
UNIT 1	**SECTION 2**
5	An airline booking platform assists passengers in selecting the most suitable travel itinerary from multiple available flight options. Although all itineraries reach the same destination, they differ in ticket price, layover duration, total travel time, and the number of connecting flights. The platform evaluates these alternatives based on the passenger's preferences and travel constraints to recommend the most appropriate option. Assess the design of path-cost functions for selecting an appropriate itinerary and supporting decision-making. (5 Marks)
6	A mobile robot performs inventory inspection inside a large warehouse. During the search process, it repeatedly revisits the same storage locations through different movement sequences, increasing travel time and battery consumption.

	Outline an appropriate strategy for avoiding repeated states and justify its impact on search efficiency. (5 Marks)
7	A National Disaster Response Force (NDRF) team deploys an autonomous rescue robot inside a partially collapsed underground mining tunnel after an earthquake. The exact location and depth of trapped miners are unknown, and the robot has only a limited battery and onboard memory. The rescue operation must ensure that no reachable survivor is overlooked while avoiding excessive memory consumption during the search. Conclude the most suitable uninformed search strategy for this rescue mission and justify the recommendation based on the characteristics of the search problem. (5 Marks)
8	(i) Consider the tree with 16 nodes given in the figure below.  When Iterative deepening search is to be used, predict the output of each iteration for the above tree. Also, indicate all the iterations in which node 8 is reached. (3 Marks)
	(ii) Demonstrate Depth first search, to reach all the nodes in the tree. (2 Marks)
UNIT 2	**SECTION 1**
9	An autonomous underwater vehicle (AUV) uses the A* search algorithm to inspect underwater pipelines.Before deployment, engineers evaluate the heuristic function by comparing the estimated remaining cost with the actual remaining cost from different pipeline inspection points.

Inspection point	Actual Remaining Cost	Estimated Cost
A	22	20
B	18	16
C	15	18
D	9	8
Goal	0	0

	Formulate the heuristic values and interpret whether the heuristic function is suitable for deployment in the AUV navigation system.Justify your recommendation. (5 Marks)
10	A logistics company develops an AI-based warehouse navigation system using the A* search algorithm to retrieve products efficiently. During performance testing, the engineers estimate that the heuristic function has an effective branching factor of 2, and the optimal solution is found at a search depth of 5. The effective branching factor is given by: $$N + 1 = 1 + b^* + (b^*)^2 + \cdots + (b^*)^d$$

	where: N = Total number of nodes generated b ∗ = Effective branching factor d = Solution depth Compute the total number of nodes generated by the search algorithm using the above equation and assess what the computed value indicates about the computational effort required to reach the solution. (5 Marks)
11	A cloud service provider uses a Genetic Algorithm to optimize the allocation of virtual machines to physical servers. Each chromosome represents a possible resource allocation strategy. Figure shows one generation of the optimization process.  Fig: Optimization Process Apply the sequence of operations shown in the figure to generate a new population of resource allocation strategies. (5 Marks)
12	(i) An autonomous inspection robot in a manufacturing plant represents component positions using an 8-puzzle model. Before selecting the next search node, the robot estimates the remaining effort using two heuristic functions. The initial and goal configurations are shown below.  Fig. 8 Puzzle model Compute the heuristic estimates obtained using the Misplaced Tiles heuristic and the Manhattan Distance heuristic. (3 Marks)
	(ii) Find the best heuristic that provides a more informative estimate for the given puzzle configuration.Justify. (2 Marks)
UNIT 2	**SECTION 2**
13	A smart home energy management system schedules household appliances.Each appliance has its own operating-time restrictions (unary constraints), and some appliances cannot run simultaneously due to power capacity limits (binary constraints).Before scheduling, the system removes invalid time slots and incompatible appliance combinations. Analyze the effect of enforcing Node Consistency followed by Arc Consistency on the scheduling process before the search begins.

		(5 Marks)
14	A cloud gaming company generates Sudoku puzzles for millions of users every day. The puzzle generator must fill the empty cells while ensuring that each row, each column, and every 2×2 subgrid contains the digits 1 to 4 exactly once. The system verifies every assignment before filling the next empty cell and revises previous assignments whenever a conflict is detected. Consider Figure a: Initial state of the sudoku puzzle and Figure b: Partial solution for the sudoku puzzle.  Figure a: Initial state of the sudoku puzzle  Figure b: Partial solution for the sudoku puzzle Carry-out the Backtracking Search algorithm to Figure b and generate a valid Sudoku solution satisfying all constraints. Also identify the number and its positions that caused conflict in the configuration given in figure b. (5 Marks)	
15	A smart manufacturing system assigns robots to production operations.After each assignment, incompatible robot choices are removed from the remaining operations.If any operation has no feasible robot, the scheduler backtracks and tries another assignment. Predict the CSP inference technique adopted by the manufacturing controller to complete the scheduling process and justify its operations. (5 Marks)	
16	(i) A semiconductor fabrication facility schedules wafer processing through multiple processing chambers. During scheduling, the controller discovers that a resource conflict originates from an earlier chamber assignment rather than from the most recent allocation. Instead of revisiting every previous assignment, the scheduler directly returns to the assignment responsible for the conflict and continues the search from that point. Breakdown the intelligent backtracking strategy employed by the scheduler by organizing the sequence of conflict identification and backtracking. (3 Marks)	
	(ii) Justify the effectiveness of intelligent backtracking in reducing unnecessary search operations compared with chronological backtracking. (2 Marks)	
UNIT 3	**SECTION 1**	
17	A university management system stores academic policies using First-Order Logic. The system represents rules such as "Every registered student can access the digital library" using universal quantifiers before verifying student access requests. Interpret the purpose of using universal quantifiers in representing academic policies within the university management system.	

(5 Marks)

18 A smart hospital stores the rule "Every patient diagnosed with diabetes receives dietary counselling." During system testing, the AI engineer notices that the inference engine is unable to identify patients who should receive counselling because the rule has not been represented correctly in First-Order Logic.

Assess the given rule and formulate an appropriate First-Order Logic expression using suitable predicates and quantifiers.Justify your answer.

(5 Marks)

19 A smart city management system represents information from multiple city services using structured knowledge by identifying objects, relations, functions, and properties from natural language statements.

(a) "Electric Bus B12 transports 45 passengers from Central Bus Terminal to City Hospital in 18 minutes."

(b) "Traffic Signal T8 gives priority to Ambulance A15 during emergency operations at Junction J4."

(c) "Air Quality Sensor S21 measures a PM2.5 concentration of 32 µg/m³ in Industrial Zone Z5."

Identify the objects, relations, functions (if any), and properties (if any) represented in the statements and complete the table given below.

Statements	Objects	Relations	Functions	Properties
a				
b				
c				

(5 Marks)

20 (i) A smart warehouse management system automates inventory movement using autonomous robots. The system maintains information about robots, containers, storage zones, and transportation tasks in a First-Order Logic knowledge base to support automated decision-making. Before assigning a transportation task, the system represents individual facts as atomic sentences and operational rules as complex sentences to enable logical reasoning.

Formulate the atomic sentence for the following facts.

(a) Robot R12 carries Container C15.

(b) Container C15 is stored in Zone Z4.

(c) Robot R12 is active.

(3 Marks)

(ii) Interpret the following requirement and represent it as a complex sentence.

"If Robot R12 is active and Container C15 is available, then Robot R12 transports Container C15."

(2 Marks)